

Sets

- **Definition 1**

A **set** is a collection of objects, called **elements** of the set. If x is an element in a set A we denote it $x \in A$, and $x \notin A$ otherwise.

- **Example 1.1**

If we have 3 pens in a pencil case x_1, x_2, x_3 , then

$$B = \{x_1, x_2, x_3\}.$$

And if we consider the blue pens in a pencil case

$$C = \{x \in B : x \text{ is blue}\}.$$

For example, if x_1 and x_3 are blue pens we have that

$$C = \{x_1, x_3\}.$$

- **Definition 2**

Let A be a set. We say that a set B is a **subset** of A , denoted $B \subseteq A$, if all the elements in B are also elements in A , that is, if $x \in B$ then $x \in A$ for all $x \in B$.

- **Definition 3**

Let A and B be sets. We say that $A = B$, if $A \subseteq B$ and $B \subseteq A$.

- **Definition 4**

If $B \subseteq A$ and $A \neq B$, we say that B is a **proper subset** of A and is denoted $B \subsetneq A$ (or $B \subset A$).

- **Example 4.1**

Which of the following is a set?

$$\begin{array}{ll} (a) \{6, 7, 8, 9, 10\} & (b) \{\{e, d, f\}, g, h\} \\ (c) \{e, d, f, g, h\} & (d) \{6, 7, \{8, 9\}, 10\} \end{array}$$

a, b, c, d are sets.

- **Example 4.2**

Note that

$$A = \{6, 7, \{8, 9\}, 10\}$$

Determine if the following expressions are valid.

$$\begin{array}{ll} 6 \in A \checkmark & \{7, \{8, 9\}\} \subseteq A \checkmark \\ 8 \in A \times & \{8, 10\} \subseteq A \times \\ 8 \in A \times & \{\{8, 9\}, 10\} \in A \times \\ \{8, 9\} \in A \checkmark & \end{array}$$

- **Definition 5**

The **empty set**, denoted $\emptyset$, is the set that has no elements.

- **Remark 5**

Let A be a set, then $\emptyset \subseteq A$ and $A \subseteq A$.

- **Definition 6**

Let A and B be two sets.

The **union** of A and B , denoted $A \cup B$, is the set that contains all elements of A and all elements in B , that is,

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

The **intersection** of A and B , denoted $A \cap B$, is the set that contains all elements of A that are also elements of B , that is,

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

The **difference** A minus B , denoted $A \setminus B$, is the set that contains all elements of A that are not in B , that is,

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}.$$

- **Example 6.1**

Consider the sets $A = \{a, b, c, d, e\}$ and $B = \{b, c\}$, then

$$A \cup B = \{a, b, c, d, e\}$$

$$A \cap B = \{b, c\}$$

$$A \setminus B = \{a, d, e\}$$

$$B \setminus A = \emptyset = \{ \}.$$

- **Definition 7**

Let X be a set and consider two subsets A_1 and A_2 of the set X .

We say that A_1 and A_2 are **disjoint** if $A_1 \cap A_2 = \emptyset$.

We say that A_1 and A_2 form a **partition** of X if $X = A_1 \cup A_2$ and $A_1 \cap A_2 = \emptyset$ with $A_1, A_2 \neq \emptyset$.

- **Definition 8**

Let A be a set, the **complement** of A , denoted A^c , is the set $A^c = \{x : x \notin A\}$.

- **Proposition 8.1**

Let X be a set.

1. If $A, B \subseteq X$ then $A, B \subseteq A \cup B$ and $A \cap B \subseteq A, B$.
2. If $A, B \subseteq X$ then $A \cup B = B \cup A$ and $A \cap B = B \cap A$.
3. If $A, B \subseteq X$ and $A \subseteq B$ then $B^c \subseteq A^c$.
4. $X^c = \emptyset$ and $\emptyset^c = X$.
5. If $A, B \subseteq X$ then $(A \cup B)^c = A^c \cap B^c$.
6. If $A, B \subseteq X$ then $(A \cap B)^c = A^c \cup B^c$.

5, 6 are De Morgan's laws.

Proof

1. Let $x \in A$, then $x \in A$ or $x \in B$, that is,

$$x \in A \cup B \implies A \subseteq A \cup B.$$

Analogously, we have that $B \subseteq A \cup B$.

Let $x \in A \cap B$, then $x \in A$ and $x \in B$, that is,

$$x \in A \cap B \implies A \cap B \subseteq A, B.$$

- 2.

$$\begin{aligned} A \cup B &= \{x : x \in A \text{ or } x \in B\} \\ &= \{x : x \in B \text{ or } x \in A\} = B \cup A. \end{aligned}$$

Analogously, we have that $A \cap B = B \cap A$.

3. Suppose that $A \subseteq B$ and let $x \in B^c$, then $x \notin B$; since $A \subseteq B$ we have that $x \notin A$ and thus $x \in A^c$.

Therefore, $B^c \subseteq A^c$.

4. By definition:

$$X^c = \{x \in X : x \notin X\} = \emptyset$$

$$\emptyset^c = \{x \in X : x \notin \emptyset\} = X.$$

5. Let $x \in (A \cup B)^c$, then

$$\begin{aligned} x \in (A \cup B)^c &\iff x \notin (A \cup B) \\ &\iff \text{It is false that } x \in A \vee x \in B \\ &\iff \text{It is true that } x \notin A \wedge x \notin B \\ &\iff x \notin A \text{ and } x \notin B \\ &\iff x \in A^c \text{ and } x \in B^c \\ &\iff x \in A^c \cap B^c. \end{aligned}$$

Thus $(A \cup B)^c = A^c \cap B^c$.

6. Let $x \in (A \cap B)^c$, then

$$\begin{aligned} x \in (A \cap B)^c &\iff x \notin (A \cap B) \\ &\iff \text{It is false that } x \in A \wedge x \in B \\ &\iff \text{It is true that } x \notin A \vee x \notin B \\ &\iff x \notin A \text{ or } x \notin B \\ &\iff x \in A^c \text{ or } x \in B^c \\ &\iff x \in A^c \cup B^c. \end{aligned}$$

Thus $(A \cap B)^c = A^c \cup B^c$.

- **Definition 9**

Let X be a set. The **power set** of X , denoted $\mathcal{P}(X)$, is the set of all subsets of X .

- **Example 9.1**

Let $X = \{a, b, c\}$, then

$$\mathcal{P}(X) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}.$$

- **Example 9.2**

Let A and B be sets. Prove that $\mathcal{P}(A) \subseteq \mathcal{P}(B)$, if and only if, $A \subseteq B$.

- **Proof 9.2.1**

Suppose that $\mathcal{P}(A) \subseteq \mathcal{P}(B)$. Let $x \in A$ then $\{x\} \in \mathcal{P}(A)$ and thus $\{x\} \in \mathcal{P}(B)$, therefore $x \in B$, hence $A \subseteq B$.

Suppose that $A \subseteq B$. Let $X \in \mathcal{P}(A)$ then $X \subseteq A$, that is, $X \subseteq B$, therefore $X \in \mathcal{P}(B)$ and thus $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.